

Gerard D. Ompad

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SUMMARY

Working for more than 15 years, I have an extensive expertise in computational statistics, data science and artificial intelligence, with a proven track record in both technical and leadership roles. Highly skilled in managing cross functional teams and delivering innovative A.I. and data driven solutions across diverse industries, including healthcare, pharmaceuticals, retail, energy, and telecommunications.

INDUSTRY EXPERIENCE

Statistical Data Science Manager, Pfizer Inc. Jan 2026 - Present

Involved in the ideation, conception, prototyping of statistical programs for clinical trials. Involved in the creation of agentic models and A.I. frameworks to assist in clinical studies and reporting. Mentoring a group of statistical data scientists in developing programs on different therapeutic agents under clinical investigation.

Senior Principal Data Scientist, DTN MeteoGroup Oct 2020 - Jan 2026

Involved in the ideation, conception, prototyping, and development of new technologies for fuels, and environment monitoring. Utilizing multi-modal and spatial data to create new products powered by artificial intelligence and data science.

Customers and Insights Manager, Globe Telecom Nov 2019 - Oct 2020

Leads a team of data scientists and data engineers to analyze customer behavior, and generate Know-Your-Customer report. Utilizing big data to investigate customer habits, experiences and decisions to create descriptive personas to assist in the development of new telco products.

Data Science Manager, Tiaxa Sep 2018 - Nov 2019

Leads a team of data scientists, data engineers, and business analysts to develop new products for financial inclusivity by analyzing mobile usage, previous lending patterns, and current behavior of principal borrowers in Chile, Philippines, Nepal, and Bangladesh.

Senior Machine Learning Engineer, Natural Intelligence Solutions Jun 2015 - Aug 2018

Develop machine learning algorithms and pipelines to cater to local industries in Philippines, Singapore, and Malaysia. Initiates the conception, planning and prototyping of minimum viable products to assist customers in their data science and machine learning requirements and needs.

Analytics Consultant, United Laboratories Jun 2014 - Jun 2015

Leads in the development of analytics dashboard for pharmaceutical supply chain and sales.

Biostatistician, Globetekpro Science Foundation Jan 2012 - Dec 2013

Conducts analysis of patient data undergoing chemotherapy and dendritic cell therapy.

Technical Consultant, Youth Lead Dec 2010 - Jan 2012

Develops survey methodologies and performs analysis for UNICEF and UNFPA funded projects focusing on young key affected people.

Statistician, The Global Fund Oct 2010 - Jan 2011

Assisting stakeholders and sub-recipients in the development of survey methodologies and analysis of HIV epidemiological data.

ACADEMIC AND RESEARCH EXPERIENCE

Lecturer, Department of Statistics, University of the Philippine-Cebu, 2025-Present

Teaches probability theory and statistics to undergraduate computer science students.

Lecturer, Department of Computer Science, University of Cebu, 2025-Present

Teaches python programming and fundamental of artificial intelligence to undergraduate computer science students.

Researcher, Data and Implementation Sciences for Health, 2025-Present

Assists in the research and development of new health technologies for tuberculosis, and HIV.

Researcher, Drug Safety Group, University of Copenhagen, 2024-Present

Assist in the conception, ideation, prototyping and development of generative artificial intelligence for drug safety, pharmacovigilance, and pharmacoepidemiology. Supports graduate students in developing codes and algorithms for their thesis and dissertations.

Lecturer, Department of Computer Science, University of San Carlos, 2023-2025

Teaches linear algebra, fundamentals of data analytics, and fundamental of artificial intelligence to undergraduate computer science students.

EDUCATION

2017 - present MSc. (Computational) Statistics at **University of the Philippines Diliman**

2011 - 2017 MSc. Tropical Medicine at **University of the East Ramon Magsaysay Memorial Medical Center, Inc.**

2003 - 2009 BSc. Chemical Engineering at **University of San Carlos**

PUBLICATIONS

Antepuesto, Erwin et al. (2024). "Investigating Causal Relationships Between Inflation News Among Other News Topics in Philippine News Media Using Granger Causality". In: *2024 IEEE International Conference on Communication, Networks and Satellite (COMNETSAT)*. IEEE, pp. 152–160. ISBN: 979-8-3503-6808-6. DOI: [10.1109/COMNETSAT63286.2024.10862725](https://doi.org/10.1109/COMNETSAT63286.2024.10862725). URL: <https://ieeexplore.ieee.org/document/10862725>.

Bacus, James Vincent et al. (2024). "Detecting Diseases in Corn through Convolutional Neural Network Architectures and Ensemble". In: *Proceedings of the 2024 7th Artificial Intelligence and Cloud Computing Conference*. AICCC '24. Tokyo, Japan: Association for Computing Machinery, pp. 173–179. ISBN: 9798400717925. DOI: [10.1145/3719384.3719409](https://doi.org/10.1145/3719384.3719409). URL: <https://doi.org/10.1145/3719384.3719409>.

Reponte, Ronan Jasper G. et al. (2024). "Multi-Network Based Approach for Drug Repurposing". In: *2024 IEEE International Conference on Communication, Networks and Satellite (COMNETSAT)*. IEEE, pp. 405–412. ISBN: 979-8-3503-6808-6. DOI: [10.1109/COMNETSAT63286.2024.10862464](https://doi.org/10.1109/COMNETSAT63286.2024.10862464). URL: <https://ieeexplore.ieee.org/document/10862464>.

Abainza, Alexandra R. et al. (2025). "De Novo Design of Gram-Negative Antibacterial Compounds Using Junction-Tree Variational Autoencoders and Computational Screening". In: *2025 IEEE Asia-Pacific Conference on Geoscience, Electronics and Remote Sensing Technology (AGERS)*. IEEE, pp. 158–165. ISBN: 979-8-3315-8789-5. DOI: [10.1109/AGERS67633.2025.11446444](https://doi.org/10.1109/AGERS67633.2025.11446444). URL: <https://doi.org/10.1109/AGERS67633.2025.11446444>.

- Adriatico, Isabel Joy D. et al. (2025). "Deep Learning Methods to Predict Sea Surface Temperature and Marine Heatwave Occurrence in the Philippine Sea". In: *Software Engineering: Emerging Trends and Practices in System Development. CSOC 2025*. Ed. by Radek Silhavy and Petr Silhavy. Vol. 1560. Lecture Notes in Networks and Systems. Springer, Cham, pp. 164–175. DOI: [10.1007/978-3-032-00239-6_11](https://doi.org/10.1007/978-3-032-00239-6_11). URL: https://link.springer.com/chapter/10.1007/978-3-032-00239-6_11.
- Elderfield, Zeus D. et al. (2025). "Deep Learning-Based Modeling of Marine Heatwave Events: Identifying Key Exogenous Drivers and Enhancing Predictive Accuracy". In: *2025 IEEE Asia-Pacific Conference on Geoscience, Electronics and Remote Sensing Technology (AGERS)*. IEEE, pp. 499–506. ISBN: 979-8-3315-8789-5. DOI: [10.1109/AGERS67633.2025.11446353](https://doi.org/10.1109/AGERS67633.2025.11446353). URL: <https://doi.org/10.1109/AGERS67633.2025.11446353>.
- Manalili, Marlex Lance et al. (2025). "Extension of the Sessa Empirical Estimator to Clustering Techniques with Non-Convex Assumptions". In: *2025 IEEE International Conference on Communication, Networks and Satellite (COMNETSAT)*. Padang, Indonesia: IEEE, pp. 36–42. DOI: [10.1109/COMNETSAT68601.2025.11324906](https://doi.org/10.1109/COMNETSAT68601.2025.11324906). URL: <https://doi.org/10.1109/COMNETSAT68601.2025.11324906>.
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- Ompad, Gerard D. et al. (2025). "Off-the-Shelf Large Language Models for Guiding Pharmacoepidemiological Study Design". In: *Clinical Pharmacology & Therapeutics*. Early View. DOI: [10.1002/cpt.70039](https://doi.org/10.1002/cpt.70039). URL: <https://ascpt.onlinelibrary.wiley.com/doi/10.1002/cpt.70039>.
- Pasco, Jericho et al. (2025). "Utilizing Deep Learning to Predict the Potency of Beta-Lactamase Inhibitors". In: *2025 13th International Conference on Bioinformatics and Computational Biology (ICBCB)*. Cebu City, Philippines: IEEE, pp. 103–110. DOI: [10.1109/ICBCB64873.2025.11198104](https://doi.org/10.1109/ICBCB64873.2025.11198104). URL: <https://ieeexplore.ieee.org/document/11198104>.
- Sanchez, Bryan Ignatius P. et al. (2025). "Spatiotemporal and Causal Analysis of Dengue Transmission in Cebu Province, Philippines". In: *2025 IEEE International Conference on Communication, Networks and Satellite (COMNETSAT)*. Padang, Indonesia: IEEE, pp. 16–22. ISBN: 979-8-3315-8256-2. DOI: [10.1109/COMNETSAT68601.2025.11324640](https://doi.org/10.1109/COMNETSAT68601.2025.11324640). URL: <https://doi.org/10.1109/COMNETSAT68601.2025.11324640>.
- Heckmann, Nicole Sonne et al. (2026). "Optimizing Large Language Models for Causality Assessment in Pharmacovigilance: Developing a Performance Metric as Objective for Bayesian Hyperparameter Optimization". In: *Arxiv*. DOI: <https://doi.org/10.48550/arXiv.2607.03704>. URL: <https://doi.org/10.48550/arXiv.2607.03704>.
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- Rossi, Andrea et al. (Sept. 2026). "The clinical burden of potentially inappropriate medications on hospitalization risk in an Italian older adult cohort: a large-scale real-world study". In: *International Journal of Pharmacy Practice*, riag112. DOI: [10.1093/ijpp/riag112](https://doi.org/10.1093/ijpp/riag112). URL: <https://doi.org/10.1093/ijpp/riag112>.

SKILLS AND TOOLSETS

Pytorch	Main library used to create deep learning prototypes.
Python	Main programming language used fo general purpose tasks.
Plotly	Main library used for visualization and front end web application and development.
R	Main programming language used for statistics specific research projects.
SAS	Main programming language used for highly confidential patient data.
Bash	Main programming language used for Linux based tasks.
SQL & PostGresSQL	Main programming language used for data storage and management.
AWS	Main cloud base platform to store data online (S3 Buckets and Glacier Buckets).
Microsoft Office	Main tools used for administrative based documentation.
Latex	Main tools used for research documentation and journal publication.
Atlassian	Main tool used for technical documentation and task monitoring (Confluence, JIRA).
Github	Main tool used for version control and code repository.